

NITIK JAIN

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Graduate student at Johns Hopkins University majoring in **Robotics**, with 3+ years of early-stage startup experience building production AI systems across healthcare AI, autonomous vehicles, and fintech.

Seeking **Robotics Engineer, Research Engineer, Computer Vision, Perception, and Applied ML roles.**

EDUCATION

Johns Hopkins University

Aug 2025 – May 2027

MSE Robotics (Perception & Cognitive Systems)

Baltimore, MD

- J. N. Tata Scholar & K. C. Mahindra Scholar 2025 amongst ~100 scholars selected worldwide for postgraduate studies

Indian Institute of Technology (IIT), Kanpur

Jul 2017 – May 2022

Integrated B.Tech + M.Tech, Mechanical & Aerospace Engineering · Minor in Machine Learning · GPA 9.4/10

Kanpur, India

- Academic Excellence Award 2021 — top academic performance across the dual-degree program

PUBLICATIONS & PATENTS

- Multimodal Object Detection Under Sparse Forest-Canopy Occlusion  Preprint (extends M.Tech thesis), 2026
- Computer-Implemented Stigma Separation System  Indian Patent No. 564565, 2025
- IR Models and the COVID-19 Pandemic: A Comparative Study  Europe PMC, 2023
- Development of a Low-Cost Autonomous Ground Vehicle  IEEE ICARSC, 2022
- Lightweight Multi-Drone Detection and 3D-Localization via YOLO  arXiv, 2022

WORK EXPERIENCE

Lead AI Engineer, Swiffylabs.ai

Jun 2024 – Jun 2025

WestBridge Capital-backed B2B fintech; owned AI product + engineering for a 0→1 core-banking platform

Bangalore, India

- Led AI product and engineering for a 0→1 core-banking platform, translating NBFC workflows into modular services
- Shipped LLM-agent and fraud-detection automation across onboarding, transactions, and customer-experience workflows
- Drove commercial go-live of a regulatory-ready banking suite across **3 international markets**, launching in **50% less time** than the legacy industry standards

Founding AI Engineer, Osfin.ai

Apr 2023 – Jun 2024

Sequoia-backed fintech building AI-first financial automation

Bangalore, India

- Owned AI product + engineering go-live for a large private bank reconciliation suite processing **4M+ transactions/day**
- Enabled OCR for multilingual unstructured invoice text, improving extraction and matching accuracy by **40%**
- Shipped exception-management workflows that cut recon TAT by **4x**, freed finance-ops teams from manual review, and prevented **400k** in financial leakage at that client

AI Scientist, R&D Division, Qure.ai

Sept 2022 – Mar 2023

3D medical-vision R&D for qCT-Chest, Qure.ai's flagship CT-based lung-health AI suite

Bangalore, India

- Built 3D vision models for CT lung-nodule classification, achieving **88% AUC / 96% recall** on malignancy prediction
- Improved nodule segmentation using attention-augmented U-Net models with CBAM + SE blocks, increasing DICE by **14%**
- Advanced qCT workflows for nodule detection, quantification, and follow-up, supporting deployments processing **1,200 CTs/week** across **4 NHS Trusts**

AI Research Consultant, MooVita Pte Ltd

Sept 2022 – Feb 2023

Computer-vision perception and explainability for autonomous driving

Singapore

- Estimated **3D sparse point cloud** by triangulating 2D lane features through monocular visual odometry
- Engineered a CPU-only **Lucas-Kanade optical-flow** fallback for turn-rate estimation, running at **240 FPS**

RESEARCH EXPERIENCE

Student Researcher, Johns Hopkins Medicine

Aug 2025 – Present

Deep learning for coronary-centerline localization in Philips scout cardiac MRI Advisor: Dr. Mike Schär

Baltimore, MD

- Building an automated coronary artery localization pipeline from Philips cardiac MRI and expert point annotations
- Developed a **2.5D ResNet50+FPN** model with 5-slice context for per slice coronary centerpoint detection
- Built a full scanner-side cardiac-MRI app for dataset ingestion, coronary label review, and automated scan processing

Sr. Student Researcher, IGCL, IIT Kanpur

Jun 2021 – Aug 2022

M.Tech Thesis: Through-canopy object detection using RGB-thermal fusion Advisor: Dr. Mangal Kothari

Kanpur, India

- Developed a through canopy perception pipeline combining **RGB thermal fusion**, LiDAR feasibility tests, and swarm system

- Used **Airborne Optical Sectioning** synthetic aperture imaging to suppress canopy clutter and recover occluded targets
 - Fine-tuned **YOLOv5** on fused thermal imagery, removing **~70%** occlusion and improving detection accuracy by **32%**
- Visiting Research Intern, RRC, IIIT Hyderabad** **May 2019 – Jul 2019**
- Outdoor Visual Inertial State Estimation and Motion Planning* Advisor: Prof. K. Madhava Krishna Hyderabad, India
- Integrated **VINS-Mono** in ROS with hardware-synced camera, IMU, and **GPS-RTK** for drift-robust UAV odometry
 - Built a RealSense-depth voxel mapper and connected it to `mav_voxblox_planning` for real-time 3D path planning

SELECTED PROJECTS

PAVE: Probing Affordance in Vision-Language-Action Encoders 🔗

Studying whether robot policy VLA models preserve object affordance and its impact on downstream tasks

- Evaluated whether robot-policy models retain object-affordance signals after end-to-end fine-tuning
- Showed frozen **DINOv2-large** features reach **0.776 mIoU** on UMD Part-Affordance, outperforming SoTA by **10.6 pp**
- Found VLA fine-tuning causes class-specific affordance loss, with some action categories degrading far more than others
- Built an adapter that recovered **82%** of lost affordance performance, showing feature drift can be partially corrected

Daksh II: Autonomous Ground Vehicle 🔗

Team VISiON, IIT Kanpur, outdoor self-driving research platform for IGVC autonomy challenges

- Built an end-to-end **ROS autonomy stack** for outdoor grassy-terrain navigation with lane keeping and obstacle avoidance
- Implemented **Fast-SCNN** semantic segmentation using MobileNet-style blocks for low-latency lane perception
- Developed LiDAR-camera occupancy mapping and integrated **costmap + RRT** planning for autonomous navigation

Extreme-Illumination Image Restoration 🔗

Restoring usable visual detail from low-light and multi-exposure RGB images

- Built a **U-Net** restoration model for severely exposed images, targeting detail recovery under extreme illumination
- Evaluated restoration quality using PSNR and SSIM, reaching **18.44 dB PSNR** and **0.782 SSIM** on benchmark images
- Outperformed Zero-DCE++ by **6.95 dB**, showing stronger reconstruction quality with a compact image-restoration pipeline

GENIE: Latent Structure with Diffusion Models 🔗

ML4SCI project on generative modeling for quark and gluon jet calorimeter images

- Built a **VAE-to-diffusion** pipeline to compress, reconstruct, and generate jet images across CERNs calorimeter dataset
- Achieved **37.93 dB** reconstruction quality and **0.967 SSIM**, showing the latent representation preserved jet-image structure
- Validated learned representations with a **GraphSAGE** classifier reaching **ROC-AUC 0.774** on downstream jet classification

AWARDS & ACHIEVEMENTS

- **J. N. Tata Scholar** and **K. C. Mahindra Scholar**, among **~100 scholars globally** selected for postgraduate funding 2025
- **SIIC Student Innovation Award**, IIT Kanpur, recognizing the top graduating student for entrepreneurship 2022
- **Academic Excellence Award**, IIT Kanpur, for institute-wide top performance in the dual-degree program 2021
- **Lescoe Cup**, 2nd worldwide, 27th Intelligent Ground Vehicle Competition, Michigan, USA 2019
- **3rd AutoNav**, 27th Intelligent Ground Vehicle Competition, among 37 international teams 2019
- **Silver Medal**, 7th Inter-IIT Tech Meet, most innovative robotic automation in agriculture to alleviate drudgery 2019
- **Bronze Medal**, 6th Inter-IIT TCS ATC Challenge, for Autonomous Lavatory Sanitation Robot 2018

LEADERSHIP, ENTREPRENEURSHIP & SERVICE

- **Team Leader, Team VISiON, IIT Kanpur**: led a 21-member autonomous-vehicle team, raised **\$35,000** and an NVIDIA grant, and helped build an IGVC-winning outdoor robotics platform 2019–21
- **Co-Founder, Tralee.ai / SkipQ**: built an AI retail checkout-automation MVP, raised **\$12K** in pre-seed funding, and reached the national finals among **1,600+** startup teams 2020–21
- **Teaching Assistant**: mentored students in Innovative Data Strategy for Global Leaders at JHU and AE640A Autonomous Systems at IIT Kanpur, spanning AI strategy, robotics, and autonomy coursework 2022, 2026
- **Academic Service**: delivered an invited talk on ML and Computer Vision at IIT Kanpur's QIP Short-Term Course and reviewed papers for the Seventh Indian Control Conference 2020–22

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB, SQL, Bash

Machine Learning: PyTorch, PyTorch Lightning, Hydra, MONAI, Hugging Face, Scikit-learn, NumPy

Computer Vision: OpenCV, 2D/3D vision, segmentation, object detection, image restoration, medical imaging

Robotics: ROS/ROS2, Gazebo, MoveIt, visual odometry, sensor fusion, occupancy mapping, motion planning

Tools: Claude Code, Cursor, Docker, Git, CMake, colcon, Databricks, Grafana, CloudWatch, ~~TeX~~